

Pregnancy Support System

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Pregnancy is among the most health-critical periods in a woman's life, yet access to continuous, personalised health monitoring remains limited for the majority of expectant mothers in Sri Lanka and across the developing world. Routine antenatal visits provide only periodic snapshots of maternal health, leaving significant temporal gaps during which dangerous complications such as gestational hypertension, gestational diabetes, or foetal distress may develop undetected until they escalate into emergencies.

This research addresses that gap by designing and implementing an AI-powered maternal health analytics and risk classification module that forms a core component of a broader four-part mobile application for pregnant women. The module accepts structured health metric inputs from the pregnant woman, including heart rate averages, blood pressure readings, and blood glucose levels, integrates them with CTG scan analysis output produced by the companion Part 1 module (IT22272386), and applies a trained multi-class risk classification model to produce an actionable risk level: Low, Medium, High, or Very High. When the risk level reaches High or Very High, the system automatically notifies the attending doctor through the doctor-facing application interface, attaching the full set of health metrics and the classification rationale.

A second key feature of this module is a daily AI-driven health questionnaire. Each day, the system presents the pregnant woman with a personalised set of symptom-check questions covering pain, foetal movement, sleep quality, swelling, and general wellbeing. Her responses are stored and a longitudinal behavioural analysis model tracks deviations from her personal baseline over time. When the response pattern indicates an emerging health concern, an early alert is issued to the woman. If the concern reaches a high-risk threshold, a notification is automatically sent to the doctor.

The expected outcomes include a classification model achieving above ninety percent accuracy on a labelled maternal health dataset, automated doctor notification triggered correctly for all High and Very High risk classifications in testing, behavioural anomaly detection issuing early warnings at least twenty-four hours before a simulated health deterioration event, and a daily questionnaire completion rate above eighty percent in user testing. This module directly integrates with the CTG scan analysis component (IT22272386) as a data consumer and feeds risk-level outputs to the broader application's doctor notification and consultation management layers.

Keywords: *Maternal Health, Pregnancy Risk Classification, AI Health Monitoring, Gestational Diabetes, Hypertension Detection, Behavioural Pattern Analysis, Machine Learning, Antenatal Care, Sri Lanka*

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LIST OF ABBREVIATIONS

ADC	Analog to Digital Converter
AI	Artificial Intelligence
API	Application Programming Interface
BP	Blood Pressure
CNN	Convolutional Neural Network
CTG	Cardiotocography
EHR	Electronic Health Record
FCM	Firebase Cloud Messaging
GDM	Gestational Diabetes Mellitus
HR	Heart Rate
IoT	Internet of Things
JSON	JavaScript Object Notation
ML	Machine Learning
NLP	Natural Language Processing
REST	Representational State Transfer
RF	Random Forest
SDK	Software Development Kit
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology

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1. INTRODUCTION

1.1 Background and Context

Pregnancy is a physiologically complex and health-critical period during which a woman's body undergoes rapid, continuous change. Complications such as gestational hypertension, gestational diabetes mellitus (GDM), and pre-eclampsia are among the leading causes of maternal and foetal morbidity and mortality worldwide, and Sri Lanka is no exception. According to the World Health Organization, over seventy percent of maternal deaths occur due to conditions that are preventable with adequate monitoring and timely medical intervention [1]. The challenge is not the absence of medical knowledge; it is the absence of continuous, accessible health monitoring between antenatal visits.

In Sri Lanka, the standard antenatal care protocol involves scheduled clinic visits at intervals of four to eight weeks, with more frequent visits only in the later stages of pregnancy or when a complication is already identified. Between these visits, changes in the woman's heart rate, blood pressure, or blood sugar level that may signal an emerging complication go undetected. A woman experiencing a gradual rise in blood pressure over a week is unlikely to recognise it as significant until it becomes symptomatic, by which point the condition may have already progressed to a dangerous stage [2].

Cardiotocography (CTG) scanning is a standard monitoring tool used to assess foetal wellbeing through heart rate and uterine contraction patterns. While CTG results carry significant diagnostic value, their interpretation traditionally requires specialist expertise, and the results are communicated to patients in a clinical shorthand that most pregnant women cannot interpret independently. This creates an information asymmetry that reduces the woman's ability to participate in her own health management [3].

Mobile health technology offers a practical pathway to close this monitoring gap. Smartphones are widely owned across Sri Lanka, including in rural areas, and a well-designed mobile application can serve as a continuous health monitoring companion between clinic visits. Recent advances in machine learning, particularly ensemble classifiers and gradient boosting models trained on maternal health datasets, have demonstrated strong predictive accuracy for pregnancy risk classification from structured health metrics, making AI-assisted risk assessment a clinically viable capability for mobile deployment [4].

This research develops the AI-powered health analytics and risk classification module of a four-part maternal health mobile application. The module accepts structured health metric inputs from the pregnant woman; integrates them with CTG scan analysis output from the companion Part 1 module (IT22272386); applies a trained classification model to assign a risk level of Low, Medium, High, or Very High; and triggers automated doctor notification when the risk level reaches a clinically significant threshold. A parallel daily health questionnaire feature tracks the woman's symptom responses over time, building a longitudinal behavioural profile that enables early anomaly detection before risk metrics escalate to critical levels [5].

1.2 Literature Review

1.2.1 AI-Based Maternal Health Risk Assessment

The application of machine learning to maternal health risk prediction has attracted substantial research attention over the past decade. Ahmed et al. (2020) conducted a systematic review of AI-based maternal mortality prediction models, finding that ensemble methods including Random Forest and Gradient Boosting consistently outperformed logistic regression baselines on structured clinical datasets, achieving AUC scores above 0.88 across multiple study populations [4]. Their review identified blood pressure, blood glucose, and heart rate as the three most predictive individual features for risk classification, which directly informs the selection of health metrics collected by this module.

Sufian et al. (2022) developed a pregnancy risk classification system using a dataset of maternal health records from rural Bangladesh, applying a Random Forest classifier to distinguish between low, medium, and high risk categories from six clinical features. Their model achieved ninety-one percent accuracy on the test set, and their analysis confirmed that the combination of blood pressure and blood sugar together produced significantly better classification accuracy than either feature alone [5]. This finding supports the multi-metric input design adopted in this research, where all available health metrics are provided to the classifier simultaneously rather than treating each in isolation.

1.2.2 CTG Scan Interpretation and AI Analysis

Ayres-de-Campos et al. (2015) provided a comprehensive review of automated CTG analysis systems, noting that the major limitation of manual CTG interpretation is high inter-observer variability among clinicians. AI-based CTG analysis systems trained on large annotated datasets have been shown to reduce this variability and flag non-reassuring traces that human observers may miss during busy clinical sessions [6]. The present research does not itself implement CTG analysis, which is handled by the Part 1 module (IT22272386), but the integration of CTG outputs into the risk classification input vector as a supplementary feature is a key design decision supported by this evidence. A combined health-metrics-plus-CTG classifier is expected to outperform a metrics-only classifier.

1.2.3 Health Metric Monitoring and Disease Prediction in Pregnancy

Dua et al. (2020) compared seven machine learning classifiers for high-risk pregnancy prediction using a dataset of over five thousand antenatal records, finding that XGBoost and Random Forest produced the highest F1-scores (0.91 and 0.89 respectively) and were the most robust to class imbalance, which is a common issue in maternal risk datasets where high-risk cases are under-represented [10]. Their feature importance analysis consistently ranked diastolic blood pressure, fasting blood glucose, and gestational week as the top three predictors, providing strong empirical justification for including these exact parameters in the health metric input form of this module.

Bharti et al. (2021) further validated the use of ensemble classifiers for cardiac event risk prediction in pregnant populations, demonstrating that combining heart rate variability features with blood pressure readings substantially improved classification recall for high-risk cases [11]. This result motivates the inclusion of heart rate average as a first-class input feature rather than treating it as supplementary.

1.2.4 Behavioural Pattern Analysis and Anomaly Detection in mHealth

Rajpurkar et al. (2022) demonstrated that longitudinal health tracking models trained on sequential patient symptom responses significantly outperform single-point prediction models for chronic condition deterioration detection. Their key finding, that deviations from an individual's personal baseline are more predictive than absolute values alone, is the theoretical foundation for the daily questionnaire behavioural analysis feature in this research [7]. The implication is that the system should track each user's personal response history rather than applying a single population-level threshold.

Liaw and Liu (2019) explored smartphone-based daily symptom tracking for obstetric care in low-resource settings, finding that daily micro-surveys of five to eight questions achieved completion rates above eighty-three percent when questions were brief, contextually relevant, and accompanied by immediate personalised feedback [8]. The daily questionnaire design in this research applies these principles directly, keeping each group focused and providing instant feedback after submission.

1.2.5 Existing Products and Systems

Several commercial and open-source options exist for pregnancy health monitoring, but none combine the cost profile, multi-metric AI classification capability, automated clinical alerting, and longitudinal behavioural analysis that this research targets:

- Ovia Pregnancy: A widely used pregnancy tracking app with basic symptom logging and content recommendations. It does not perform AI-based multi-metric risk classification and has no automated doctor notification mechanism. It operates entirely as a consumer wellness tool rather than a clinical decision support tool.
- Flo Health: A menstrual and pregnancy cycle tracking application with AI-based insights. Its pregnancy tracking features are primarily informational and do not integrate structured clinical metrics such as blood pressure or blood glucose for risk classification. No automated clinical alert functionality exists.
- WHO ANC Application: A reference implementation of WHO antenatal care guidelines in mobile form. It provides educational content and appointment scheduling guidance but does not perform personalised risk classification from user-entered health metrics or provide automated doctor alerting.
- Hospital-grade EHR mobile modules: Enterprise systems used in private hospitals that include some remote monitoring capability. These are expensive to deploy (typically LKR 500,000+ per clinic), require IT department integration, and are designed for clinical staff rather than the pregnant woman herself.

Capability	Ovia	Flo Health	WHO ANC App	Our System
Multi-metric AI risk classification	Partial	Partial	No	**Yes**
CTG output integration	No	No	No	**Yes**
Automated doctor notification	No	No	Partial	**Yes**

Daily behavioural pattern tracking	No	Partial	No	**Yes**
Longitudinal anomaly detection	No	No	No	**Yes**
Low-cost / free access	Partial	Partial	Yes	**Yes**

1.3 Research Gap

The literature review and product analysis above reveal four specific gaps that no existing solution fully addresses in combination:

Gap	What is Missing	Why It Matters	Our Solution
1	No single low-cost app integrates multi-metric health inputs (HR, BP, blood glucose) with CTG scan output for unified AI risk classification	Single-metric assessments miss compound risk patterns; BP and glucose together predict risk far better than either alone	ML classifier trained on all collected metrics plus CTG output from IT22272386 produces a single four-level risk score per submission
2	Existing apps lack automated doctor notification triggered by AI risk thresholds with full clinical context	Delayed clinical awareness of high-risk states is a leading cause of preventable complications escalating into emergencies	Automatic notification with full health metric context and SHAP-generated contributing factor summary dispatched to doctor's app when risk is High or Very High
3	No existing app performs longitudinal behavioural anomaly detection from daily symptom questionnaire responses against a personal baseline	Behavioural patterns often shift before clinical metrics deteriorate, offering an early warning window that is missed without daily tracking	Daily questionnaire responses are stored and analysed against the woman's personal rolling baseline; anomalies trigger early alerts before risk metrics cross thresholds
4	No existing affordable solution provides four-class risk stratification (Low / Medium / High / Very High) with explainable AI output	Binary or three-class risk models lose clinical nuance needed for appropriate triage and prioritisation of doctor attention	Four-class classification with SHAP-explainable output indicating which contributing metrics most influenced the classification result

The novelty of this research lies in bringing all four solutions together into a single deployable mobile module. The integration context further strengthens the research value: risk classification output is directly consumed by the doctor notification layer and shared with the chatbot module (IT22638168), allowing validation against real application outcomes rather than testing the classifier in isolation.

2. OBJECTIVE

2.1 Main Objective

To design and implement an AI-powered maternal health analytics and risk classification module that collects structured health metrics from pregnant women, integrates them with CTG scan analysis outputs from the companion module (IT22272386), applies a trained multi-class risk classification model to produce actionable Low / Medium / High / Very High risk assessments, automatically notifies attending doctors when risk reaches a clinically significant threshold, and tracks daily symptom questionnaire responses to detect longitudinal behavioural anomalies and issue early health warnings.

2.2 Specific Objectives

- Design and implement a health metric input interface within the pregnant woman's application that collects heart rate averages, systolic and diastolic blood pressure, and blood glucose levels, together with gestational week, and persists this data in a structured format for AI model consumption. Target: input form completable by a first-time user in under two minutes.
- Train and deploy a multi-class risk classification model on a labelled maternal health dataset that incorporates the collected health metric inputs and CTG analysis output from IT22272386 as features, producing a four-level risk label (Low / Medium / High / Very High). Target: weighted F1-score and classification accuracy above ninety percent on the held-out test set.
- Implement automated doctor notification: when the classification model outputs a High or Very High risk level, automatically generate and deliver a structured push notification to the doctor-facing application containing the full health metric set, the risk label, and a SHAP-generated contributing factor summary. Target: notification delivered to doctor device within thirty seconds of risk classification being computed.
- Design and implement the daily health questionnaire feature, presenting the pregnant woman with a personalised set of daily symptom-check questions covering pain and discomfort, foetal movement, sleep and fatigue, swelling and visual changes, and general wellbeing, storing all responses in the application database. Target: questionnaire delivery and response capture working end-to-end in the application.
- Implement a behavioural anomaly detection model that compares each day's questionnaire response vector against the woman's personal rolling baseline and flags responses that deviate significantly, issuing an in-app early alert to the woman and, at high-severity anomaly, a notification to the doctor. Target: early alert issued at least twenty-four hours before a simulated health deterioration event in controlled testing.

3. METHODOLOGY

Research Approach:

This research follows the Design Science Research (DSR) methodology, which is well suited to engineering projects that produce a working technological artefact and evaluate it against defined performance benchmarks. The five phases below map to the DSR cycle of problem identification, design, construction, evaluation, and communication.

- Phase 1 – Review maternal health AI literature; identify and compare suitable datasets and model architectures; select classifier platform; define performance targets; agree data contract and API schema with IT22272386 (CTG module).
- Phase 2 – Acquire and preprocess datasets; perform exploratory data analysis and feature engineering; train baseline models; run stratified k-fold cross-validation to select the best-performing classifier.
- Phase 3 – Implement the health metric input interface and risk classification REST API. Integrate CTG output feature from IT22272386 into the classification pipeline. Implement SHAP explainability layer.
- Phase 4 – Implement the automated doctor notification system using Firebase Cloud Messaging. Build the daily questionnaire engine and behavioural anomaly detection model.
- Phase 5 – End-to-end integration testing with all four modules. User testing with simulated health data scenarios. Performance measurement against all defined targets.

Technical Methods — Health Metrics and Data Collection:

The following health parameters are collected from the pregnant woman through the mobile application interface:

Health Parameter	Unit	Clinical Significance	Normal Range (Pregnancy)
Heart Rate (average)	bpm	Elevated HR may indicate infection, anaemia, or cardiac stress	60 – 100 bpm
Systolic Blood Pressure	mmHg	Key indicator for hypertensive disorders of pregnancy	< 140 mmHg
Diastolic Blood Pressure	mmHg	Diastolic ≥ 90 is a diagnostic threshold for hypertension in pregnancy	< 90 mmHg
Blood Glucose Level	mmol/L	GDM screening and monitoring; elevated glucose affects foetal growth	< 7.8 mmol/L (post-meal)
Gestational Week	Weeks	Risk profiles differ significantly across	1 – 42 weeks

		trimesters; used as contextual normalisation factor	
CTG Risk Score (from IT22272386)	Categorical	Foetal wellbeing indicator integrated from CTG analysis module as supplementary feature	Normal / Suspect / Pathological

Technical Methods — Risk Classification Model:

The risk classification model takes the collected health metric vector as input and outputs one of four risk levels: Low, Medium, High, or Very High. The model development follows this sequence:

1. **Dataset**: UCI Fetal Health Classification Dataset (2,126 records) and the Maternal Health Risk Dataset (1,014 records) are combined and preprocessed. Features are aligned to the parameters collected by the application interface.
2. **Model selection**: Random Forest, Gradient Boosting (XGBoost), and Support Vector Machine (SVM) classifiers are trained and compared using stratified ten-fold cross-validation. The model with the highest weighted F1-score is selected for deployment.
3. **Feature engineering**: Interaction terms between systolic and diastolic blood pressure and blood glucose are computed, as prior literature identifies combined BP-glucose patterns as particularly predictive. Gestational week is used as a contextual normalisation factor across all features.
4. **Explainability**: SHAP (SHapley Additive exPlanations) values are computed for each prediction to generate a human-readable summary of which metrics contributed most to the classification. This summary is included in the doctor notification payload alongside the risk label.
5. **Deployment**: The trained model is serialised using joblib and served via a Python/Flask REST API endpoint. The mobile application sends health metric JSON payloads to this endpoint and receives the risk level and SHAP summary in the response within the latency target.

Technical Methods — Daily Questionnaire and Behavioural Anomaly Detection:

Each day, the application presents the pregnant woman with a set of symptom-check questions. The question set is organised into five parameter groups as shown in Table 6. Response vectors for each day are stored in the application database. The behavioural anomaly detection model computes a rolling Z-score for each response item against the woman's personal historical baseline, requiring a minimum of seven days of prior data before anomaly detection activates. A composite anomaly score is computed as the weighted sum of per-item Z-scores. When the composite score exceeds a moderate threshold, an in-app early alert is displayed to the woman. When it exceeds a high-severity threshold, a doctor notification is dispatched automatically.

Question Group	Example Questions	Response Encoding
Pain and Discomfort	Stomachache today? Headache today? Back pain today?	Binary yes/no per question
Foetal Movement	Did you feel baby move today? Reduced movement compared to yesterday?	Binary + 3-point scale
Sleep and Fatigue	Approximately how many hours did you sleep last night? Feeling unusually fatigued today?	Numeric + Binary
Swelling and Visual Changes	Swelling in feet or hands today? Any blurred vision or visual changes?	Binary yes/no
General Wellbeing	How do you feel overall today? (Rate 1 – 5)	5-point Likert scale

Technical Methods — Doctor Notification System:

The doctor notification system operates as follows: whenever a risk classification of High or Very High is produced, or when the daily anomaly score exceeds the high-severity threshold, a structured notification payload is constructed. This payload contains the patient's name and ID, gestational week, all submitted health metrics, the risk level label, the SHAP-generated contributing factor summary, and a timestamp. The payload is delivered to the doctor-facing application via Firebase Cloud Messaging (FCM), which supports both Android and iOS push notification delivery. The doctor can view the full health context within the application, initiate a chat with the patient, or schedule an appointment directly from the notification detail screen.

Tools and Technologies:

Tool	Purpose	Justification
Python (scikit-learn, XGBoost)	Model training, evaluation, deployment	Industry standard for ML; extensive maternal health dataset support and community
SHAP library	Model explainability for doctor notifications	Best-established feature attribution method; produces human-readable summaries compatible with non-technical users
Python / Flask	REST API serving risk classification endpoint	Lightweight; integrates cleanly with mobile application frontend via JSON
Firebase (Firestore + FCM)	Database storage and push notification delivery	Real-time sync; FCM handles cross-platform push delivery reliably for both Android and iOS
React Native / Flutter	Mobile application frontend (coordinated with team)	Cross-platform single codebase for Android and iOS; consistent UI across all four modules

Jupyter Notebook	Model development, EDA, and evaluation	Standard interactive ML development environment; reproducible training notebooks
Multimeter / Reference instruments	Validation of health metric input accuracy against manual reference readings	Essential for verifying that user-entered values produce correct classification outcomes
Git / GitHub	Version control and team collaboration	Shared repository for model code, API schema, and integration contracts

Validation Strategy:

Test	Procedure	Success Criteria
Risk classification accuracy	Evaluate trained model on held-out 20% test split of labelled maternal dataset using stratified sampling	Weighted F1-score and accuracy above 90% on test set
Doctor notification delivery	Submit High and Very High risk inputs; timestamp at classification and at push notification receipt on doctor's device; compute difference across 100 samples	Notification delivered within 30 seconds in 95% of test cases
Behavioural anomaly detection lead time	Simulate 7-day normal baseline, then inject abnormal response pattern; verify alert timing relative to simulated deterioration event	Early alert issued at least 24 hours before simulated deterioration
CTG integration benefit	Submit classification requests with and without CTG risk score feature; compare weighted F1-scores across both conditions	CTG feature improves F1-score by minimum 3% over metrics-only baseline
Questionnaire end-to-end	Complete daily questionnaire flow on test device for 7 days; verify all responses stored correctly	Zero data loss across 7-day simulated test period; all responses retrievable for anomaly model
Retry on notification failure	Disable FCM connectivity during a High-risk classification; restore and verify notification delivered on retry	All failed notifications retried and delivered within 3 attempts

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The maternal health application is built from four integrated components, each developed by a different team member, forming a complete pipeline from clinical data input through to personalised health support and real-time clinical alerting:

Layer	Component	Member	Role
1	CTG Scan Analysis & AI Interpretation	IT22272386	Manual entry or camera-based photo capture of CTG report diagrams; AI analysis of CTG values and waveforms; result summary with identified issues, causes, and preventive measures delivered to both woman and doctor
2	**Health Analytics & Risk Classification (This Project)**	**IT22557506**	**Multi-metric health metric input; ML-based four-level risk classification; automated doctor notification with SHAP explanation; daily symptom questionnaire; longitudinal behavioural anomaly detection**
3	AI Chatbot with Mood Detection	IT22638168	NLP-based conversational chatbot; passive mood inference via front camera analysis and typing behaviour; therapeutic music and video recommendations; full chat history saved locally for offline access
4	AI Recipe and Nutrition Suggestion	IT22284716	Daily ingredient inventory capture; AI-generated pregnancy-safe complete meal recommendations from available ingredients; calorie levels and health benefit information for each suggested meal

This project (Layer 2) sits at the intersection of clinical data and automated alerting. The CTG module (Layer 1, IT22272386) produces a risk score that feeds directly into Layer 2's classification input vector, making Layer 2 partially dependent on Layer 1 for maximum classification accuracy. Layer 2 in turn produces the risk level that determines whether the doctor notification system is activated, making reliable risk classification a prerequisite for the application's core clinical safety function. The chatbot module (Layer 3, IT22638168) queries the current risk level from Layer 2 to contextualise chatbot responses appropriately for the woman's health state.

Data Flow — Health Metrics to Doctor Dashboard:

6. Pregnant woman opens the application and navigates to the health log screen, entering today's heart rate average, blood pressure (systolic and diastolic), and blood glucose reading.

7. The application retrieves the latest CTG analysis risk score from the Part 1 module's data layer (IT22272386) and appends it to the health metric payload as a supplementary feature.
8. The combined payload is transmitted to the Flask risk classification API endpoint. The trained ML model produces a four-class risk label and SHAP attribution summary identifying the top contributing features.
9. The risk level and plain-language explanation are displayed to the woman on her health dashboard with a colour-coded visual indicator (green/yellow/orange/red).
10. If risk is High or Very High, an automated notification is immediately dispatched via Firebase Cloud Messaging to the doctor's application, containing the full health metric context and SHAP summary.
11. Doctor receives the notification, views the full health summary within the doctor-facing application, and can initiate a consultation response or schedule an appointment directly.
12. Simultaneously, the daily questionnaire engine presents that day's symptom questions. Responses are stored in Firestore and fed into the behavioural anomaly detection model against the woman's personal baseline.
13. If the anomaly model detects a significant deviation, an in-app early alert is shown. At high-severity anomaly, a separate doctor notification is triggered with the flagged response pattern attached.

The API contract defining the health metric JSON payload schema and the CTG risk score integration format is agreed with IT22272386 and locked by Week 8 of the project timeline to prevent integration delays.

5. USER REQUIREMENTS

Stakeholder Identification:

Stakeholder	Interaction with This Module	Primary Need
Pregnant Woman	Primary user; inputs health metrics daily; receives risk level feedback, daily questionnaire, and anomaly early alerts	Simple input interface; understandable risk feedback in plain language; actionable early warnings before conditions worsen
Attending Doctor	Receives automated push notifications when risk reaches High or Very High; reviews full health metric context	Timely, accurate alerts with clinical context; SHAP explanation of contributing factors; minimal false positives
CTG Analysis Module (IT22272386)	Provides CTG risk score as a supplementary feature input to the risk classification model	Agreed API schema for CTG result output; stable data contract locked by Week 8
AI Chatbot Module (IT22638168)	Queries current risk level from this module to contextualise chatbot responses for the woman's health state	Accessible risk level API endpoint with consistent response format and low latency
System Administrators	Monitor classification API uptime, model performance metrics, and notification delivery logs	Logging, error reporting, and model performance dashboards accessible without deep technical setup

Functional Requirements:

ID	Requirement	Priority
FR-01	Accept structured health metric inputs (heart rate, systolic BP, diastolic BP, blood glucose, gestational week) from the pregnant woman through the mobile application interface	Critical
FR-02	Retrieve the latest CTG analysis risk score from IT22272386 and incorporate it as a feature in the risk classification input vector for every classification request	Critical
FR-03	Classify the combined health metric and CTG input into one of four risk levels — Low, Medium, High, Very High — using the trained and deployed ML model	Critical
FR-04	Display the risk level to the pregnant woman with a colour-coded visual indicator and a plain-language explanation of the contributing factors derived from SHAP values	Critical

FR-05	Automatically generate and deliver a push notification to the doctor's application when risk level is High or Very High, containing full health metric context, risk label, and SHAP contributing factor summary	Critical
FR-06	Present a daily health symptom questionnaire to the pregnant woman covering pain and discomfort, foetal movement, sleep and fatigue, swelling and visual changes, and general wellbeing question groups	Critical
FR-07	Store all daily questionnaire responses in Firestore and build a longitudinal behavioural profile per registered user	Critical
FR-08	Run behavioural anomaly detection on each new questionnaire response set; issue an in-app early alert when the composite anomaly score exceeds the moderate threshold	High
FR-09	Send an automated doctor notification when the behavioural anomaly score exceeds the high-severity threshold, attaching the flagged response pattern	High
FR-10	Expose a current risk level API endpoint accessible by the chatbot module (IT22638168) to retrieve the most recent risk classification for the logged-in user	Medium

Non-Functional Requirements:

Category	Requirement
Classification Accuracy	ML model must achieve weighted F1-score and accuracy above ninety percent on the held-out test set
Notification Latency	Doctor notification must be delivered within thirty seconds of a High or Very High risk classification being computed under stable network conditions
Anomaly Detection Lead Time	Behavioural anomaly alert must be issued at least twenty-four hours before a simulated health deterioration event in controlled testing
Data Privacy	All health metric data must be stored encrypted at rest in Firestore; all transmission between app and API must use HTTPS; no raw health data shared with third parties
Usability	Health metric input screen completable by a first-time user in under two minutes; daily questionnaire completable in under three minutes
Reliability	Classification API must maintain ninety-nine percent uptime during the test period; failed notifications must be retried automatically up to three times before logging a failure

Data Format	JSON API output must conform to the schema agreed with IT22272386 by Week 8 and must remain stable for the remainder of the project timeline

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona:

The primary market for this application module is pregnant women in Sri Lanka seeking accessible, continuous health monitoring between antenatal clinic appointments. Sri Lanka has a smartphone penetration rate above eighty percent, and the majority of pregnant women in both urban and rural settings own a device capable of running the application. Government antenatal care programmes and private obstetric clinics represent a secondary institutional market, as they can integrate the platform into their patient management workflow to improve risk monitoring coverage between scheduled visits without requiring investment in additional clinical hardware.

The attending doctor is a key secondary user persona: obstetricians and general practitioners managing antenatal care caseloads who benefit from AI-assisted triage that surfaces high-risk patients automatically rather than relying entirely on patient-initiated contact. For doctors managing large antenatal panels in district hospitals, automated risk stratification represents a genuine workload efficiency improvement as well as a clinical safety benefit.

Value Proposition:

The module delivers AI-powered, continuous maternal health risk monitoring at zero cost to the end user beyond smartphone ownership. For pregnant women, it closes the monitoring gap between clinic visits with daily health tracking, personalised risk feedback, and early warnings. For doctors, it provides automated triage of high-risk patients with full clinical context, reducing the risk of missed deterioration events and enabling proactive intervention. No existing free or low-cost application in the Sri Lankan market delivers the combination of multi-metric risk classification, CTG integration, automated clinical alerting, and longitudinal behavioural monitoring.

Competitive Advantage:

Feature	Our Module	Ovia	Flo Health	Hospital EHR
Classification accuracy	**>90% (AI model)**	N/A	N/A	N/A
Multi-metric + CTG classification	**Yes**	No	No	Partial
Automated doctor notification	**Yes**	No	No	Yes
Behavioural anomaly detection	**Yes**	No	No	No
Cost to end user	**Free**	Freemium	Freemium	LKR 500k+ (clinic)
SHAP explainable output	**Yes**	No	No	No

Revenue Model and Pricing:

- Freemium model: core health monitoring and risk classification features free for all users; premium tier includes detailed trend analytics, historical report export, and appointment scheduling integration.
- Institutional licensing: hospitals and obstetric clinics pay a per-practitioner annual subscription to access the doctor-facing application with patient panel management, bulk notification management, and EHR export APIs.
- Government health programme partnerships: Ministry of Health integration for national antenatal programmes, with subsidised or zero-cost deployment for public sector use and grant funding from SDG-aligned health technology initiatives.

Intellectual Property:

The specific combination of multi-metric maternal health classification with CTG output integration, automated SHAP-explainable doctor notification, daily behavioural anomaly detection from longitudinal questionnaire responses, and four-class risk stratification represents a practical system design not documented as a complete application in existing literature. The trained model weights and anomaly detection algorithm are candidates for academic publication and open-source release to maximise adoption in maternal health applications across developing countries while establishing this project as the reference implementation.

7. BUDGET AND JUSTIFICATION

As a software-focused component, this budget reflects cloud services, dataset access, development tools, documentation, and user testing costs:

#	Item	Total (LKR)	Justification
1	Google Firebase Spark Plan (free tier) — Firestore database and FCM push notifications	0	Free tier sufficient for prototype and all testing volumes; no cost incurred during the research period
2	Google Colab Pro subscription x3 months for GPU-accelerated model training	3,200	Standard GPU instances required for XGBoost and ensemble training on full combined dataset within reasonable time
3	Maternal Health Risk Dataset and UCI Fetal Health Dataset (open source, Kaggle/UCI)	0	Both datasets freely available under open licences; no procurement cost
4	VPS hosting for Flask API deployment during integration testing x3 months	4,500	Low-cost VPS (DigitalOcean or equivalent) required for API endpoint accessible to mobile app during team integration testing
5	Android test device for FCM notification delivery testing (shared with team)	0	Personal devices available across team members; no additional procurement required
6	User testing participant incentives (10 participants x LKR 500)	5,000	Small incentive for user testing session participants to ensure realistic engagement and usability feedback
7	Printing, binding, and documentation materials	2,000	Final report, viva poster, and project submission materials
	TOTAL	**LKR 14,700**	

All primary development tools (Python, scikit-learn, XGBoost, SHAP, Flask, Jupyter, VS Code, Git) are free and open-source. The SLIIT IT Laboratory provides development workstations, internet access, and software licences at no additional cost. Human effort is covered within the academic project framework.

8. WORK BREAKDOWN STRUCTURE (WBS)

#	Task	Member	Duration
1	Literature review, dataset identification, model architecture comparison, and API schema design with IT22272386	IT22557506	Week 1–3
2	Dataset acquisition, preprocessing, exploratory data analysis, feature engineering, and class imbalance handling	IT22557506	Week 3–4
3	Model training, hyperparameter tuning, and cross-validation (RF, XGBoost, SVM comparison)	IT22557506	Week 4–6
4	SHAP explainability integration; model serialisation and Flask API endpoint development	IT22557506	Week 6–7
5	Health metric input interface implementation in mobile application (coordinated with team)	IT22557506 + team	Week 5–7
6	CTG output integration: data contract with IT22272386; feature pipeline update and combined classifier testing	IT22557506 + team	Week 7–8
7	Doctor notification system: Firebase FCM integration; notification payload design, dispatch logic, and delivery testing	IT22557506	Week 7–9
8	Daily questionnaire engine: question presentation, response capture, Firestore storage schema	IT22557506	Week 8–9
9	Behavioural anomaly detection model: rolling Z-score implementation, baseline engine, threshold calibration, alert dispatch logic	IT22557506	Week 9–11
10	End-to-end integration testing with IT22272386 (CTG input) and IT22638168 (risk level API)	IT22557506 + team	Week 10–12
11	User testing with simulated health scenarios; usability feedback collection and interface refinement	IT22557506 + team	Week 11–13
12	Analysis, documentation, final report writing, and viva demonstration preparation	IT22557506	Week 13–14

Key Milestones

Milestone	Target Week
Dataset preprocessed; baseline model training completed with initial accuracy benchmarks	Week 4
Final ML model selected; classification accuracy target (>90%) achieved on held-out test set	Week 6
Flask API deployed; classification endpoint responding correctly to test payloads	Week 7

CTG data contract agreed and feature integration pipeline functional with IT22272386	Week 8
Doctor FCM notifications working end-to-end for High and Very High risk triggers	Week 9
Daily questionnaire and behavioural anomaly detection fully implemented and tested	Week 11
All validation tests passed; final report submitted and live demo ready for viva	Week 14

9. GANTT CHART

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
1. Literature & Dataset Design														
2. Data Preprocessing & EDA														
3. Model Training & Evaluation														
4. SHAP & Flask API														
5. Health Input Interface														
6. CTG Integration														
7. Doctor Notification System														
8. Daily Questionnaire														
9. Anomaly Detection														
10. Integration Testing														
11. User Testing														
12. Report & Demo Prep														

10. REFERENCES LIST

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11. APPENDICES

Appendix A — Risk Analysis

Risk	Likelihood	Impact	Mitigation Strategy
ML model accuracy does not reach ninety percent target on the selected dataset	Medium	High	Test multiple classifier architectures (RF, XGBoost, SVM) from the outset; expand training data by combining multiple publicly available maternal health datasets; apply SMOTE to address class imbalance if accuracy falls short
CTG output from IT22272386 not available in time for integration testing	Low	Medium	Design classification model to function with or without CTG feature; CTG is an additive input, not a prerequisite for baseline function; agree and lock data contract with IT22272386 by Week 8
FCM push notification delivery fails or is unreliable during testing	Low	High	Implement notification retry logic up to three attempts; add in-app notification centre as fallback so doctor can see missed alerts even without push delivery
Behavioural anomaly model produces excessive false-positive early alerts	Medium	Medium	Require minimum seven days of baseline before anomaly detection activates; tune composite Z-score threshold conservatively; validate against simulated deterioration scenarios before deployment to live users
Pregnant women provide inaccurate health metric inputs due to manual entry errors	High	Medium	Implement input validation with physiologically plausible ranges for each metric; flag outlier values with a user-friendly prompt requesting re-entry; log input timestamp for audit trail
Classification API server downtime during a critical monitoring window	Low	High	Cache the most recent health metric submission locally on the device if the API is unreachable; automatically process the buffered submission when connectivity is restored and notify the user
Time constraints preventing full user testing completion	Medium	Medium	Prioritise technical validation targets (model accuracy, notification latency, anomaly detection) first; user testing with simulated scenarios is supplementary validation and does not block the viva demonstration

Appendix B — Personnel and Facilities

Research Team:

Role	Student ID	Contribution
Health Analytics & Risk Classification (This Project)	IT22557506	ML risk classification model, health metric input interface, CTG feature integration, doctor notification system, daily questionnaire engine, behavioural anomaly detection
CTG Scan Analysis & AI Interpretation	IT22272386	Camera-based CTG image capture, AI CTG waveform interpretation, result summary delivery to woman and doctor including identified issues and preventive measures
AI Chatbot with Mood Detection	IT22638168	NLP-based conversational chatbot, passive facial mood inference via front camera, typing behaviour analysis for mood scoring, therapeutic music and video recommendations, local offline chat history
AI Recipe and Nutrition Suggestion	IT22284716	Daily ingredient inventory input, AI-generated pregnancy-safe complete meal recommendations from available ingredients, calorie levels and health benefits output

Facilities and Resources:

Facility	Use
SLIIT IT Laboratory	Model development, API development, integration testing, documentation, and team coordination sessions
Google Colab Pro	GPU-accelerated ML model training, hyperparameter tuning, and cross-validation across all candidate classifiers
Firebase (free Spark tier)	Firestore database for all health data and questionnaire response storage; FCM for push notification delivery to doctor devices
Personal Smartphones (Android / iOS)	End-to-end application testing and FCM notification delivery validation across device types
SLIIT Campus Network	API endpoint connectivity testing, Firebase integration validation, and team integration test sessions
Team Members' Laptops	Model development and application debugging at all times outside of laboratory hours (zero additional cost)

— End of Proposal Document —

IT22557506 — AI-Powered Maternal Health Analytics and Risk Classification System

Department of Information Technology, SLIIT — March 2026